

Ella Capellini

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RESEARCH INTERESTS

I am interested in the theoretical foundations of reliable evaluation and reasoning about learned systems, particularly where hierarchical structure, statistical dependence, and limited data complicate inference. My current research develops and formally compares hierarchical Bayesian models of neural responses to naturalistic speech. This work has involved identifying failures of standard cross-validation assumptions in temporally structured data, redesigning evaluation procedures around conditionally independent units, and validating implementations against synthetic ground truth. I am interested in extending this methodological perspective to AI safety: understanding when evaluations of complex AI systems provide reliable evidence, how their statistical assumptions can be justified, and how formal analysis can reveal failures that conventional evaluation may miss.

EDUCATION

Carl von Ossietzky University of Oldenburg
MSc in Neuroscience

09/2023 - Present
Oldenburg, Germany

Relevant Modules:

Neuroscientific Data Analysis (Python and Matlab), Computational Neuroscience (Encoding/Decoding and Biophysical Modelling), Auditory Neuroscience, Language and Hearing, Psycholinguistics, Computational Intelligence I

Catholic University of the Sacred Heart
BSc in Psychological Sciences

09/2020 - 09/2023
Milan, Italy

Thesis: “The Neurophysiology of Hallucinatory Phenomena”

Relevant Modules:

Fundamentals of Neurobiology and Genetics, Developmental Psychology, Clinical Psychology, Applied Cognitive Psychology, Neuropsychology and Anatomofunctional Base of Cognitive Processes, Psychology of Personality, Psychology of Adolescence, Methods and Techniques of Psychological Assessment and Interview, Research Methods in Psychology I and II, Psychometric Statistics

RESEARCH EXPERIENCE

Institute for the Interdisciplinary Study of Language Evolution (ISLE), UZH Zürich

- **Cross-Linguistic Generalisation of the Prosodic Hierarchy: Cortical Entrainment to Italian Speech in Italian L1 Listeners**

04/2026 - Present

Master Thesis | Prof. Dr. Martin Meyer (external, UZH) & Prof. Dr. Stefan Debener (internal, Oldenburg) | in collaboration with PD Dr. Elisa Pellegrino, UZH

Asks whether the level-selective cortical entrainment pattern established for German by Oderbolz et al. (2024) — in which phase-amplitude coupling tracks a top-down structure from intonational phrase to foot to syllable — generalises to Italian, a typologically related but constraint-divergent language. Rather than a cross-linguistic comparison design, the study constructs a parallel Italian stimulus set following the same manipulation logic as the original paradigm and tests whether the same entrainment hierarchy emerges when both stimuli and listeners are Italian L1. Italian is a theoretically motivated test case because it shares core prosodic architecture with German (free lexical stress, a full prosodic hierarchy, focus encoded through prominence) but differs in constraint ranking: German prioritises rhythmic regularity, while Italian prioritises focus-driven phrasing. A difference in entrainment profiles would therefore implicate language-specific constraint ranking in shaping the quantitative expression of an otherwise shared hierarchical mechanism. The design is purely neural (EEG, ~30 Italian L1 listeners recruited in Zurich), with audio-motor synchronisation as a modular

individual-differences predictor. Core analysis: pairwise phase consistency at theta, high delta, and low delta bands via glmertree, with PAC/DAR modelling as an extension.

LiRi Lab (Language Research Infrastructure), University of Zürich **06/2026 - Present**

Research Assistant | Dr. rer. Med Fivos Iliopoulos and Prof. Dr. Nathalie Giroud

Assisting with hyperscanning EEG experiments investigating neural synchrony during social interaction, with responsibilities spanning experiment support and EEG preprocessing. Data analysis contribution and potential co-authorship anticipated as the project progresses.

Institute of Neuroinformatics (INI), ETH Zürich

Relevant Modules currently taking at ETH Zürich:

Probabilistic Artificial Intelligence, Learning in Deep Artificial and Biological Neuronal Networks, Neural Networks and Computational Complexity, and Introduction to Neuroinformatics.

• **Hierarchical Bayesian TRF Model Comparison for Speech Encoding in iEEG** **06/2026 - Present**

Semester Research Project | Prof. Dr. Timothée Proix, Neural Dynamics Lab ETH Zürich | in collaboration with Prof. Dr. Balthasar Bickel (UZH & NCCR)

Designing and implementing a formal model comparison between two hierarchical Bayesian TRF frameworks for intracranial EEG: a raised-cosine basis-function model with patient/electrode random effects, and a hierarchical Gaussian Process model in which smoothness itself (kernel lengthscale and variance) is allowed to vary by patient, electrode, and word-level surprisal. Both are compared against standard ridge-regression mTRF via held-out predictive accuracy and WAIC/LOO across seven total model variants. Surprisal (GPT-2, word-level, extending to phoneme-level via recent phoneme-LM work) enters as a predictor of neural response to naturalistic speech (The Little Prince dataset), with an explicit focus on distinguishing gain-control from predictive-coding accounts of how prediction error reshapes the temporal response function. All code (Python: NumPy, SciPy, PyMC, MNE-Python) independently validated against synthetic ground truth at each stage before use on real data; to be released on the Neural Dynamics Lab GitLab.

• **Adaptation of Vib2Sound for Missing and Degraded Accelerometer Cues** **06/2026 - 09/2026**

Semester Research Project | Prof. Dr. Richard Hahnloser Lab (supervisors: Mai Akahoshi & Yuhang Wang) ETH Zürich

Extending Vib2Sound — a deep learning model for zebra finch vocalisation separation conditioned on body-mounted accelerometer signals — to handle cue degradation across a continuous spectrum rather than binary absence. The project implements and evaluates three complementary approaches: a cascade architecture guaranteeing the model never underperforms the audio-only baseline, continuous degradation training with realistic noise augmentation (Gaussian noise, gain reduction, band masking), and a Cross-Modal Correspondence contrastive loss to shape identity-discriminative audio representations without inference cost. Additional contribution to the BirdPark annotation pipeline.

Carl von Ossietzky University of Oldenburg

09/2023 - 06/2026

• **Bilingual Processing and Auditory Perception**

Independent Research Project | Prof. Dr. Esther Ruigendijk | Summer 2025

Investigated cognitive-auditory resource competition in bilingual processing through psychophysical threshold measurement. Programmed complete experiment in Python: noise stimuli with silent gaps, pure tones at different frequencies, adaptive staircase procedures. Three German-English bilinguals completed hearing tests after 10-minute monolingual or code-switched conversations. Gap detection thresholds increased ~35-40%, pitch discrimination ~30-40% after bilingual listening, suggesting language management reduces available auditory processing resources.

Technical implementation: NumPy/SciPy for signal processing, PsychoPy for stimulus presentation, custom adaptive algorithms, statistical analysis in Python.

• **Predictive Control in Auditory Efferent Systems**

Independent Theoretical Paper | Dr. Go Ashida | Summer 2025

Proposed a predictive control framework for understanding the Medial Olivocochlear (MOC) system. Traditional models treat MOC as a reflexive feedback system; I argued it functions as a predictive controller adjusting cochlear sensitivity based on attention, task demands, and cognitive context. Formalised using control theory (Kalman filtering, optimal control), generating testable predictions about timing and learning that distinguish predictive versus reactive mechanisms.

- **Rhythmic Typology and Information Packaging: A Constraint-Based Approach to Prosodic Phrasing in German and Italian**

Independent Seminar Paper | Prof. Dr. Marcel Schlechtweg | March 2025

Developed a formal Optimality Theory model of prosodic phrasing to explain systematic differences in how German and Italian speakers encode focus. Proposed that rhythmic typology constrains information packaging: German ranks rhythmic regularity above focus-driven phrasing, while Italian reverses this hierarchy. Generated six testable predictions spanning production, perception, L2 acquisition, and typology. Engages debates on linguistic relativity and rhythm's theoretical status in phonology

- **The Morphology of the Manarolese Dialect in Relation to Standard Italian: Documenting Selective Retention During Language Shift**

Independent Seminar Paper | Dr. Phil. Dobrinka Genevska-Hanke | January 2025

Conducted original fieldwork documenting morphological attrition in a critically endangered Ligurian dialect across three generations (ages 19-83) through semi-structured interviews. Found asymmetric retention: verbal morphology collapsed entirely in the youngest generation while phonological patterns persisted in high-frequency emotionally salient vocabulary. Interpreted findings through Montreal's Incomplete Acquisition framework and Polinsky's work on heritage grammars as systematic reorganisation rather than degraded competence.

- **Code-Switching Between Manarolese and Standard Italian: Identity Negotiation Across Three Generations**

Independent Seminar Paper | Dr. Phil. Dobrinka Genevska-Hanke | January 2025

Companion study to the above analysis code-switching patterns and language attitudes in the same speaker sample. Identified three qualitatively distinct switching profiles across generations, unconscious integration, strategic alternation, and symbolic insertion, and documented a pervasive attitude-behaviour paradox where universal attachment to dialect coexists with universal shift toward Italian. Draws on Gumperz, Myers-Scotton, and Fishman to argue the paradox reflects structural constraints rather than individual failure.

INDEPENDENT COMPUTATIONAL PROJECTS

- **Homeric Formula Deviation Detector** | Python

GitHub: <https://github.com/ellacapellini/homeric-formula-deviation>

Computational framework detecting information-theoretic signatures of narrative salience in oral poetry using surprisal analysis. Parsed 15,683 lines of Ancient Greek, extracted 581 character mentions, computed $-\log_2(P(\text{formula}|\text{context}))$ using n-gram models. High-surprisal moments (4.34-6.07 bits) clustered at narrative climaxes (deaths, armour exchanges), suggesting systematic exploitation of prediction error—parallel to efficient coding in sensory neuroscience.

Technical implementation: XML parsing (lxml), n-gram frequency modelling, information-theoretic surprisal computation, interactive D3.js visualisation. Includes automatic formula extraction, Bayesian probability models, modular architecture for extension to other oral traditions or artificial language paradigms.

- **Linear B Analysis Tool** | JavaScript/Python

GitHub: github.com/ellacapellini/Linear-B-Diachronic-Phonological-Mapper

Computational toolkit for Bronze Age Greek syllabic script analysis: transcription, morphological parsing, statistical pattern analysis. Handles syllabogram-to-phoneme mapping, morphological boundaries, frequency analysis of sign co-occurrence patterns.

Technical implementation: Modular parser for syllabic transcription, statistical analysis of diachronic phonological changes (Mycenaean to Classical Greek), sign distribution visualisation. Demonstrates computational linguistics methods for under-resourced historical languages; extensible to other ancient writing systems.

CONFERENCE ATTENDANCE

Descending Auditory Pathways: Multidisciplinary Approaches
Hanse-Wissenschaftskolleg, Institute for Advanced Study

03-05 November 2025
Delmenhorst, Germany

WORK EXPERIENCE

Research Assistant | Studio of Psychiatry and Psychotherapy of Dr. Alberto Caputo

2020-2023
Milan, Italy

Assisted with forensic psychiatric assessments for court cases. Conducted literature reviews. Managed multilingual patient communication

TECHNICAL SKILLS

- Statistical modelling and evaluation: Hierarchical Bayesian inference (PyMC, NUTS/HMC), formal model comparison (WAIC, PSIS-LOO), Gaussian Process regression, synthetic data validation and recovery testing, small-N and non-exchangeable data inference
- Programming: Python (NumPy, SciPy, PyTorch, scikit-learn, NLTK, spaCy, MNE), R, MATLAB, JavaScript
- Neuroscience methods: EEG/iEEG analysis, psychophysical testing, signal processing, acoustic analysis (Praat), neuroimaging pipelines (AFNI, FSL), experimental designs
- Languages: Italian (native), English (fluent), German (learner)

ADDITIONAL COURSES

MIT OpenCourseWare

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| • Signals and Systems + Sensory Systems | 2024 |
| • Brain Mechanisms for Hearing and Speech | 2025 |
| • AFNI Training Bootcamp | 2025 |
| • fMRI Bootcamp | 2025 |
| • Brains, Minds and Machines Summer Course | 2025 |
| • Statistics for Applications | 2025 |
| • Computational Neuroscience with Neuroblox | 2026 |
| • Introduction to Deep Learning + Hands-On Deep Learning | 2026 |